

Proving Chained SHA-256 Compressions with F2Z

Protocol Specification — application profile v1

Status: normative draft; profile SHA256-F2Z-MULTICOMP-APP-V1-0000; 17 August 2026

Scope. This profile proves a public chain of SHA-256 compression calls, including feed-forward. It commits only the base witness f , derives the virtual bit tensor h , and instantiates the eleven reductions ending in the F2Z PCS opening. SHA padding, private messages, and the internals of the named GKR and Ligerito subprotocols are outside this document.

Core-profile delta. SHA256-F2Z-VIRTUAL-CORE-V1-00000 is a reserved, non-resolving candidate for a new top-level wire/core profile. It replaces the generic PCS ProfileID in the host header and tag 0x0001, and replaces direct tag 0x0003 with tag 0x1003. It also enables the generic PCS LinCheck tags 0x4002/0x4103 and the singleton tag-0x4001 terminal gate; all other inherited byte rules remain unchanged. It must not resolve to acceptance until the virtual partial-core ledger and the integrated LinCheck/ring fixtures in G6 and G7 are frozen.

1 Constraints

This section fixes the SHA application matrices. SHA-256 word semantics are those of FIPS 180-4. The row templates and the one-compression counts are those proved in [Freigen commit 93d3c55](#); this profile additionally fixes public-word binding and the canonical row sharing for a chain of N compressions.

1.1 Statement and compact matrix geometry

The public statement contains the initial and final chaining values and all message blocks:

$$x_{\text{sha}} := (N, H_0, (B_i)_{i=0}^{N-1}, H_N), \quad H_i \in (\{0, 1\}^{32})^8, \quad B_i \in (\{0, 1\}^{32})^{16}. \quad (1)$$

Every word in [Eq. \(1\)](#) has big-endian byte encoding. The canonical `app_statement_bytes` are $\text{LE}_{32}(N)$ followed by $H_0, B_0, \dots, B_{N-1}, H_N$, four big-endian bytes per word and no inner lengths. Their exact length is $68 + 64N$ bytes; trailing bytes reject. This profile requires $1 \leq N \leq 429389$, where the upper endpoint is the largest N for which $20005N + 257 \leq 2^{33}$. Before allocating, the statement preflight computes $68 + 64N$, n_f , n_h , and n_A with checked integer arithmetic and rejects on any overflow or bound failure. Padding and private-message hashing require different application profiles.

The compact Boolean input and virtual-output lengths, excluding the affine constant coordinate, are

$$n_f := 6641N + 256, \quad n_h := 20005N + 256. \quad (2)$$

Set

$$\begin{aligned} m_f &:= \max(22, \lceil \log_2 n_f \rceil), & m_h &:= \max(22, \lceil \log_2 (n_h + 1) \rceil), \\ (t_f, s, W_{\text{PCS}}) &= (m_f - 7, 7, 1), & (t_h, s, W_{\text{PCS}}) &= (m_h - 7, 7, 1). \end{aligned} \quad (3)$$

The profile requires $22 \leq m_f \leq m_h \leq 33$. Otherwise it rejects and a different profile is required. In particular the virtual chunk width is $127 - t_h - W_{\text{PCS}} = 133 - m_h \geq 100$, so $L = 1$ and $\mathcal{A} = \{0\}$. The packed-opening domain exponent is $m_p = t_f$.

Let $f \in \mathbb{F}_2^{n_f}$ be ordered as follows: all public message bits, the initial-state bits, and then the hinted decomposition bits in circuit execution order. Within a word, bit $j = 0$ is least significant and

$$[X] := \sum_{j=0}^{31} 2^j X[j], \quad \text{lo}_{32}(Q)[j] := Q[j] \quad (0 \leq j < 32). \quad (4)$$

The execution order within each compression is: rounds $0, \dots, 15$; schedule-plus-round steps $16, \dots, 61$; inlined round 62; and the fused round-63 feed-forward. Adjacent compressions reuse the previous output bits as the next state rather than allocating a duplicate state.

Let $D_f \in \mathbb{F}_2^{2^{t_f} \times 2^s}$ be the committed tensor and let J_f inject the first n_f row-major coordinates into that tensor, so its transpose selects them and $f = J_f^T \text{vec}(D_f)$. The

remaining committed padding is ignored by this relation; an honest encoder zeros it. The inherited PCS packing order is the transpose of this semantic order. Define

$$\begin{aligned} B_f[c, b] &:= D_f[b, c], \\ [\Pi_f \text{vec}(D_f)]_{c2^{t_f}+b} &:= [\text{vec}(D_f)]_{b2^s+c}, \quad \text{vec}(B_f) = \Pi_f \text{vec}(D_f), \end{aligned} \quad (5)$$

for $0 \leq b < 2^{t_f}$ and $0 \leq c < 2^s$. Thus the committed object is

$$O := \text{Enc}_{\mathcal{C}}(\text{Pack}_{128}(B_f)), \quad \rho := \text{Root}(O). \quad (6)$$

Equation (5) is the only semantic-to-PCS index permutation. P8/V8 applies it to the adjoint coefficient, and P10/V10 uses the resulting PCS order for the ring switch.

1.2 The binary matrix M_{SHA}

Use augmented vectors $\bar{f} = (1, f)$ and $\bar{h} = (1, h_{\text{sem}})$. The Freigen matrix is the fixed binary matrix

$$M_{\text{SHA}} \in \mathbb{F}_2^{(n_h+1) \times (n_f+1)}, \quad \bar{h} = M_{\text{SHA}} \bar{f}. \quad (7)$$

Let J_h inject $\bar{h}$ into the first $n_h + 1$ coordinates of $D_h \in \mathbb{F}_2^{2^{t_h} \times 2^s}$. Split $M_{\text{SHA}} = [m_0 \mid M_1]$ into its constant and nonconstant columns. The PCS form is

$$\text{vec}(D_h) = h_0 + T \text{vec}(D_f), \quad h_0 := J_h m_0, \quad T := J_h M_1 J_f^T. \quad (8)$$

Only D_f is committed. Neither M_{SHA} nor T is materialized: their rows are emitted from the following parity templates.

For a word X and $0 \leq j < 32$, define

$$\begin{aligned} \Sigma_0(X)[j] &= X[j+2] \oplus X[j+13] \oplus X[j+22], \\ \Sigma_1(X)[j] &= X[j+6] \oplus X[j+11] \oplus X[j+25], \\ \sigma_0(X)[j] &= X[j+7] \oplus X[j+18] \oplus X[j+3], \\ \sigma_1(X)[j] &= X[j+17] \oplus X[j+19] \oplus X[j+10], \end{aligned} \quad (9)$$

where every rotation index is modulo 32, while the final σ_0, σ_1 shift term is zero when $j+3 \geq 32$ or $j+10 \geq 32$, respectively. The other nonlinear-function characters are

$$P_{ef} = e \oplus f, \quad P_{eg} = e \oplus g, \quad P_{abc} = a \oplus b \oplus c. \quad (10)$$

The integer combinations used below are consequently

$$C_h := [f] + [g] - [P_{ef}] + [P_{eg}] = 2 \text{Ch}(e, f, g), \quad C_m := [a] + [b] + [c] - [P_{abc}] = 2 \text{Maj}(a, b, c). \quad (11)$$

The remaining rows of M_{SHA} are identity rows for input and hinted bits. A 34-bit schedule result derives bit zero as the XOR of its four addends' low bits and commits only bits $1, \dots, 33$; a 33-bit ordinary feed-forward result does the same for its two addends. All bits of the Q^e and Q^a decompositions are committed. Concretely, for $16 \leq k \leq 61$ and $j \in \{1, 2, 3, 5, 6, 7\}$,

$$\begin{aligned} Q_k^W[0] &= W_{k-16}[0] \oplus \sigma_0(W_{k-15})[0] \oplus W_{k-7}[0] \oplus \sigma_1(W_{k-2})[0], \\ Q_j^{\text{out}}[0] &= H_i[j][0] \oplus S_{64}[j][0]. \end{aligned} \quad (12)$$

The two identities in Eq. (12) are M_{SHA} rows, not R1CS rows. In canonical row order the exact census is

row family in M_{SHA}	rows	
affine constant	1	
message inputs; initial state	$512N + 256$	
$\Sigma_0, \Sigma_1, P_{ef}, P_{eg}, P_{abc}$ in 64 rounds	$10240N$	
σ_0, σ_1 in 48 schedule steps	$3072N$	
$Q_{16}^W, \dots, Q_{61}^W$	$1564N$	
Q^e decompositions	$2242N$	
Q^a decompositions	$2177N$	
six remaining feed-forward decompositions	$198N$	
total	$20005N + 257$	(13)

The augmented column count, in the same constant-first convention, is

$$1 + (512N + 256) + 1518N + 2242N + 2177N + 192N = 6641N + 257. \quad (14)$$

For $N = 1$, Eqs. (13) and (14) give exactly 20262 rows and 6898 columns. The previous 20457×7145 system loses 64 rows/columns by coupling a^+ to e^+ , 66 by inlining W_{62}, W_{63} , 65 by fusing final feed-forward, and another 52 columns by deriving low sum bits.

1.3 The linear $\mathbb{Z}$ -R1CS rows

For compression i , let $(a_k, b_k, c_k, d_k, e_k, f_k, g_k, h_k)$ denote the state before round k , and let W_k denote its schedule word. For $0 \leq k \leq 61$, the two round rows are

$$\begin{aligned} 2[Q_k^e] &= 2([d_k] + [h_k] + [\Sigma_1(e_k)] + K_k + [W_k]) + C_h(k), \\ 2[Q_k^a] &= 2([e_{k+1}] + [\Sigma_0(a_k)] - [d_k]) + C_m(k) + 2^{33}, \end{aligned} \quad (15)$$

where Q_k^e is 35 bits, Q_k^a is 34 bits, $e_{k+1} = \text{lo}_{32}(Q_k^e)$, and $a_{k+1} = \text{lo}_{32}(Q_k^a)$. The other six words are wiring aliases, not R1CS rows: $(b_{k+1}, c_{k+1}, d_{k+1}, f_{k+1}, g_{k+1}, h_{k+1}) = (a_k, b_k, c_k, e_k, f_k, g_k)$. Equation (15) uses $a^+ = e^+ - d + \Sigma_0(a) + \text{Maj}(a, b, c) \pmod{2^{32}}$; it does not allocate a separate T_1 or 35th Q^a bit.

For $16 \leq k \leq 61$, one schedule row is also present:

$$[Q_k^W] = [W_{k-16}] + [\sigma_0(W_{k-15})] + [W_{k-7}] + [\sigma_1(W_{k-2})], \quad W_k = \text{lo}_{32}(Q_k^W), \quad (16)$$

where Q_k^W is 34 bits and its low bit is derived as specified above. Define the unmaterialized terminal sums

$$S_k := [W_{k-16}] + [\sigma_0(W_{k-15})] + [W_{k-7}] + [\sigma_1(W_{k-2})] \quad (k \in \{62, 63\}). \quad (17)$$

Round 62 has only

$$\begin{aligned} 2[Q_{62}^e] &= 2([d_{62}] + [h_{62}] + [\Sigma_1(e_{62})] + K_{62} + S_{62}) + C_h(62), \\ 2[Q_{62}^a] &= 2([e_{63}] + [\Sigma_0(a_{62})] - [d_{62}]) + C_m(62) + 2^{33}, \end{aligned} \quad (18)$$

with 36-bit Q_{62}^e and 34-bit Q_{62}^a .

Round 63 is fused with the first and fifth feed-forward words. If $A_{\text{in}} = H_i[0]$ and $E_{\text{in}} = H_i[4]$, its rows are

$$\begin{aligned} 2[Q_e^{\text{out}}] &= 2([d_{63}] + [h_{63}] + [\Sigma_1(e_{63})] + K_{63} + S_{63} + E_{\text{in}}) + C_h(63), \\ 2[Q_a^{\text{out}}] &= 2([\text{lo}_{32}(Q_e^{\text{out}})] + [\Sigma_0(a_{63})] + A_{\text{in}} - E_{\text{in}} - [d_{63}]) + C_m(63) + 2^{34}. \end{aligned} \quad (19)$$

Here Q_e^{out} is 36 bits and Q_a^{out} is 35 bits; their low 32 bits are $H_{i+1}[4]$ and $H_{i+1}[0]$. For $j \in \{1, 2, 3, 5, 6, 7\}$, the six remaining rows are

$$[Q_j^{\text{out}}] = H_i[j] + S_{64}[j], \quad H_{i+1}[j] = \text{lo}_{32}(Q_j^{\text{out}}), \quad (20)$$

where S_{64} is the final pre-feed-forward state and each Q_j^{out} is 33 bits with a derived low bit.

The internal row count per compression is therefore

$$62 \cdot 2 + 46 + 2 + 2 + 6 = 180. \quad (21)$$

This application additionally binds the sixteen public message words per compression and the eight words of each public endpoint. Intermediate states are shared. Set $n_A := 196N + 16$ and $d_A := \lceil \log_2 n_A \rceil$. With $h := \text{lift}_2(\text{vec}(D_h)) \in \mathbb{Z}^{2^{m_h}}$, the row-padded integer constraint matrix is

$$\begin{aligned} A_x &\in \mathbb{Z}^{2^{d_A} \times 2^{m_h}}, \quad A_x h = 0, \\ A_x[\beta, *] &= 0 \quad (n_A \leq \beta < 2^{d_A}). \end{aligned} \quad (22)$$

Rows are ordered blockwise as Eqs. (15), (16) and (18) to (20), followed by message pins and the two endpoint-pin families. Omitted columns of A_x are zero; an implementation stores row templates, not the padded matrix.

Finally, projection to the fixed PCS field is accepted only after the verifier recomputes

$$B_x := \max_{\beta < 2^{d_A}} \sum_z |A_x[\beta, z]|, \quad 2B_x < 2^{39} < q = 2^{100} - 15. \quad (23)$$

Equation (23) makes reduction modulo q injective on every reachable bit-witness residual. Failure is a hard reject.

2 Prover–verifier interaction

This section instantiates the relation reductions as matched prover/verifier moves. A **NARG** record is canonical-read and immediately absorbed; a **HINT** record is never absorbed and remains pending until its predicate succeeds; a squeeze is a derived transcript event. Every NARG message below must be absorbed before the challenge that depends on it, and no squeeze is permitted while a Hint is pending. The two proof streams and all primitive encodings are those of the F2Z PCS v1 profile. The sole NARG exception is a PCS PoW nonce: parse it, permit no intervening event, verify its predicate, and only then absorb it. For the two SHA application squeeze tags, a field scalar is sampled by requesting 16 bytes with auxiliary payload $\text{LE}_{32}(a)$ for attempts $a = 0, 1, \dots$, masking the high 28 bits of the little-endian integer, and accepting the first result below q ; every attempt remains in the transcript. The transcript transition commits to each raw 16-byte XOF result before the verifier masks the local candidate. Rejection occurs before attempt 2^{32} ; the u_{32} counter never wraps.

P0/V0 — Fix the input and bind the commitment

P0. Derive $h = \text{lift}_2(h_0 + T \text{vec}(\pi_2(D_f)))$; encode and Merkle-commit only D_f , publishing ρ .

V0. Validate the public SHA chain, both tensor layouts, the application, candidate virtual-core, GKR and code profile identifiers, and ρ .

The initial obligations are

$$A_x h = 0 \text{ in } \mathbb{Z}, \quad \pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f)), \quad \rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f))). \quad (24)$$

Here T is the statement-derived SHA tap matrix and h_0 its affine column. Tag 0x0001 binds SHA256-F2Z-VIRTUAL-CORE-V1-00000, while tag 0x0002 binds SHA256-F2Z-MULTICOMP-APP-V1-0000, the subordinate identifiers, the statement digest, shape, generator, and root; V0 absorbs them in that order before any challenge. Equation (24) fixes the three claims carried by the ledger; publishing a root alone does not discharge C .

Step 1. P1/V1 — Multilinearize the constraint rows

P1. Send nothing.

V1. Pad the live rows canonically and derive their multilinear residual Q .

The exact rewrite is

$$Q(Z) := \sum_{\beta \in \{0,1\}^{d_A}} \text{eq}(\beta, Z) \langle A_x[\beta, *], h \rangle_{\mathbb{Z}}, \quad A_x h = 0 \iff Q|_{\{0,1\}^{d_A}} = 0. \quad (25)$$

Equation (25) is deterministic: P1/V1 has no message, absorb, challenge, or soundness loss.

Step 2. P2/V2 — Project the residuals into $\mathbb{F}_q$

P2. Send nothing.

V2. Set $\bar{A} = \pi_q(A_x)$ and check the statement-derived residual bound before continuing.

The mandatory gate is

$$q > 2 \max_{\beta \in \{0,1\}^{d_A}} \sum_j |A_x[\beta, j]|. \quad (26)$$

Under Equation (26), projection is injective on every reachable bit-witness residual and loses no soundness; otherwise V2 rejects.

Step 3. P3/V3 — Collapse the public constraint rows

P3. Send nothing.

V3. Squeeze $r_{\text{con}} \in \mathbb{F}_q^{d_A}$ and compute the public collapsed coefficient.

For coordinate $i = 0, \dots, d_A - 1$, V3 uses tag 0x1001, scope $(\perp_{32}, \perp_{32}, i)$, and the application field sampler above. It then derives

$$v := \bar{A}^T \text{eq}(\cdot, r_{\text{con}}), \quad E_0 : \langle v, \pi_q(h) \rangle_{\mathbb{F}_q} = 0. \quad (27)$$

OPEN CLAIMS

- S** ^{NEW}
 $A_x h = 0 \text{ over } \mathbb{Z}$
- H** ^{NEW}
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C** ^{NEW}
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$

OPEN CLAIMS

- S** ^{REDUCED}
 $Q(\beta) = 0 \text{ on } \{0,1\}^{d_A}$
- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$

OPEN CLAIMS

- S** ^{REDUCED}
 $\bar{A} \pi_q(h) = 0$
- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$

OPEN CLAIMS

- S** ^{DISCHARGED}
 $\bar{A} \pi_q(h) = 0$
- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$
- E₀** ^{NEW}
 $\langle v, \pi_q(h) \rangle = 0$

The first squeeze occurs only after V2. Equation (27) has error at most d_A/q .

Step 4. P4/V4 — Shape the claim by sumcheck

P4. For $i = 0, \dots, d_h - 1$, send the degree-two polynomial g_i under NARG tag $0x1002$ and scope $(\perp_{32}, \perp_{32}, i)$. Put $h_q := \pi_q(h)$. The tag- $0x1002$ payload is exactly the three canonical values $g_i(0), g_i(1), g_i(2)$, without a vector prefix. Starting at $\sigma_{-1} = 0$, the round and terminal rules are

V4. Absorb g_i , check its Boolean sum, then squeeze $r_i \in \mathbb{F}_q$ under tag $0x1101$ at the same scope.

$$\begin{aligned} g_i(0) + g_i(1) &= \sigma_{i-1}, & \sigma_i &= g_i(r_i), \\ r_h &= (r_0, \dots, r_{d_h-1}), & \sigma_{-1} &= 0, \quad \tau = \sigma_{d_h-1}, \quad c = \tilde{v}(r_h). \end{aligned} \quad (28)$$

Here $d_h = t_h + s = m_h$. Thus Equation (28) leaves $E_{eq} : \tilde{ch}_q(r_h) = \tau$, with error at most $2d_h/q$. Writing $r_h = (r_{row}, r_{col}) \in \mathbb{F}_q^{t_h} \times \mathbb{F}_q^s$, V4 derives $w_b = \text{can}_q(\text{eq}(b, r_{row}))$, $w'_c = \text{eq}(c, r_{col})$, and $y = \tau$. It then absorbs derived tag $0x1003$, at scope $(\perp_{32}, \perp_{32}, \perp_{32})$, with payload

$$\text{LE}_{32}(t_h) \parallel \text{LE}_{32}(s) \parallel \text{Vec}_{\mathbb{F}_q}(r_h) \parallel \text{Enc}_{\mathbb{F}_q}(c) \parallel \text{Enc}_{\mathbb{F}_q}(\tau). \quad (29)$$

The vector in Equation (29) has its canonical little-endian u_{32} count and coordinates in transcript order. Tag $0x1003$ replaces the direct profile's mandatory $0x0003/\text{CoreSeamV1}$ and must precede the first tag- $0x2001$ record. This replacement is legal only under $\text{SHA256-F2Z-VIRTUAL-CORE-V1-00000}$; the direct or an unknown core profile rejects before this event. The printed identifier is only a candidate namespace: until G6 and G7 freeze both ledgers, it must not resolve to an accepting core profile.

Step 5. P5/V5 — Expose the integer column folds

P5. Compute u_c and $\nu_c = g^{u_c}$ for every virtual column; send ν as NARG $0x2001$, then u as Hint $0x2081$. For the profile's single active chunk $\ell = 0$, the required checks are

V5. Absorb ν ; range-check and authenticate the pending u Hint, then check its reconstruction of E_{eq} .

$$u_c = \sum_b w_b D_h[b, c], \quad \nu_c = g^{u_c}, \quad \sum_c w'_c \pi_q(u_c) = y, \quad 0 \leq u_c < 2^{127}. \quad (30)$$

V5 checks every verifier-visible equality in Equation (30), including $g^u = \nu$ and the full-order condition on g , before clearing the Hint. No squeeze may occur earlier. Both vectors have exactly $2^s = 128$ entries and scope $(0, \perp_{32}, \perp_{32})$. The remaining first equality is the claim reduced by P6–P7.

Step 6. P6/V6 — Move the folds into products

P6. Send no additional field.

V6. Derive the leaf family, then squeeze the initial forest point only after V5 has cleared its Hint.

The leaf and initial claim are

$$L[b, c] := 1 + (g^{w_b} - 1)D_h[b, c], \quad \prod_b L[b, c] = \nu_c, \quad e_0 := \tilde{v}(\zeta). \quad (31)$$

V6 obtains $\zeta \in \mathbb{F}_{2^{128}}^s$ with the inherited $\mathbb{F}_{2^{128}}$ sampler and tag $0x2101$, scope $(0, \perp_{32}, i)$, in increasing $0 \leq i < s$. The exponent range makes the first reduction in Equation (31) exact; the random column compression is charged to the partial-core error.

Step 7. P7/V7 — Reduce the merged product forest to LIN

P7. Stream the selected GKR profile's sumcheck and child messages.

V7. Check and absorb every message before its challenge, returning one general linear claim on the virtual table.

OPEN CLAIMS

- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$
- E₀** DISCHARGED
 $\langle v, \pi_q(h) \rangle = \emptyset$
- E_{eq}** NEW
 $\tilde{ch}_q(r_h) = \tau$

OPEN CLAIMS

- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$
- E_{eq}** DISCHARGED
 $\tilde{ch}_q(r_h) = \tau$
- T** NEW
 $u_c = \langle w, D_h[*], c \rangle_Z$

OPEN CLAIMS

- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$
- T** DISCHARGED
 $w_c = \langle w, D_h[*], c \rangle_Z$
- P** NEW
 $\prod_b L[b, c] = \nu_c$

OPEN CLAIMS

- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$
- P** DISCHARGED
 $\prod_b L[b, c] = \nu_c$
- E_H** NEW
 $\langle R, h_{\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}} = e$

The one core-fixed GKR event is NARG tag 0x3001: the phase- c claimed sum is absorbed after the phase-link check and before the first phase- c round. The selected GKR profile owns NARG tags 0x3002–0x30ff and squeeze tags 0x3100–0x31ff; tag 0x3081 is forbidden. Its required successor is defined using the coefficientwise $\mathbb{F}_{2^{128}}$ -embeddings $f_{\mathbb{F}_{2^{128}}}, h_{\mathbb{F}_{2^{128}}}, h_{0,\mathbb{F}_{2^{128}}}$ and the scalar extension $T_{\mathbb{F}_{2^{128}}}$:

$$E_H : \quad \langle R, h_{\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}} = e, \quad h_{\mathbb{F}_{2^{128}}} = h_{0,\mathbb{F}_{2^{128}}} + T_{\mathbb{F}_{2^{128}}} f_{\mathbb{F}_{2^{128}}}, \quad (32)$$

where R is a verifier-derived coefficient function, not a transmitted dense vector. **G6** remains the requirement that the statement-bound virtual-GKR identifier publish the exact P5–P7 ledger, reduction proof, and numeric error. An unknown ledger rejects.

Step 8. P8/V8 — Apply the full virtual adjoint

P8. Send nothing.

V8. Apply the statement-bound adjoint and affine correction without a message, absorb, or squeeze.

Let $F := \Pi_f f_{\mathbb{F}_{2^{128}}}$ be the committed bits in PCS (c, b) order. V8 derives

$$a := \Pi_f T_{\mathbb{F}_{2^{128}}}^* R, \quad \mu_f := e - \langle R, h_{0,\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}}, \quad E_f : \langle a, F \rangle_{\mathbb{F}_{2^{128}}} = \mu_f. \quad (33)$$

The coefficient need not be materialized. At any LinCheck point r , V8 evaluates it by duality:

$$\tilde{a}(r) = \left\langle R, T_{\mathbb{F}_{2^{128}}} \left(\Pi_f^{-1} \text{eq}(\cdot, r) \right) \right\rangle_{\mathbb{F}_{2^{128}}}. \quad (34)$$

Equations (33) and (34) are exact. A sparse tap evaluator is conforming only if it is proved identical to this full adjoint; a columnwise shortcut rejects.

Step 9. P9/V9 — Run exactly one LinCheck

P9. For $i = 0, \dots, m_f - 1$, send one degree-two polynomial g_i .

V9. Absorb and check g_i , squeeze $r_i \in \mathbb{F}_{2^{128}}$, and carry the terminal equality to P10/V10.

Starting with $\sigma_{-1} := \mu_f$, the exact round rule is

$$g_i(0) + g_i(1) = \sigma_{i-1}, \quad r_i \leftarrow \mathbb{F}_{2^{128}}, \quad \sigma_i := g_i(r_i). \quad (35)$$

Equation (35) fixes the running claim before every challenge. Writing $g_i(X) = \gamma_{i,0} + \gamma_{i,1}X + \gamma_{i,2}X^2$, the NARG tag is 0x4002, with payload $\text{Enc}_{\mathbb{F}_{2^{128}}}(\gamma_{i,0}) \parallel \text{Enc}_{\mathbb{F}_{2^{128}}}(\gamma_{i,1}) \parallel \text{Enc}_{\mathbb{F}_{2^{128}}}(\gamma_{i,2})$ and scope $(\perp_{32}, \perp_{32}, i)$. After checking it, V9 squeezes tag 0x4103 at the same scope. The final point $r = (r_{\text{lo}}, r_{\text{hi}}) \in \mathbb{F}_{2^{128}}^7 \times \mathbb{F}_{2^{128}}^{tf}$ leaves

$$\sigma_{m_f-1} = \tilde{a}(r) \tilde{F}(r). \quad (36)$$

P10/V10 checks Equation (36) with the first ring vector. This is the only LinCheck, uses no division, and has error at most $2m_f/|\mathbb{F}_{2^{128}}|$.

Step 10. P10/V10 — Discharge LinCheck and ring-switch

P10. Send the inherited ring vector s , then retain the derived packed claim.

V10. Use s to check the LinCheck terminal, then reuse it for the inherited ring switch and singleton batch.

For $0 \leq v < 128$, P10 sends under NARG tag 0x4001, scope $(0, \perp_{32}, \perp_{32})$,

$$s_v := \widetilde{(B_f)_{v,\mathbb{F}_{2^{128}}}}(r_{\text{hi}}), \quad z := \sum_{v=0}^{127} \text{eq}(v, r_{\text{lo}}) s_v = \tilde{F}(r). \quad (37)$$

After deriving z in Equation (37), V10 absorbs s and requires $\sigma_{m_f-1} = \tilde{a}(r)z$ using Equation (34), and only then squeezes the seven tag-0x4101 coordinates of r'' . It derives (B_0, β_0) by the PCS ring-switch equation, squeezes the singleton tag-0x4102 coefficient η_0 , and sets

$$B := \eta_0 B_0, \quad \beta := \eta_0 \beta_0, \quad E_P : \langle B, P \rangle_{\mathbb{F}_{2^{128}}} = \beta, \quad P = \text{Pack}_{128}(B_f). \quad (38)$$

OPEN CLAIMS

H DISCHARGED
 $\pi_2(h) = h_0 + \text{Tree}(\pi_2(D_f))$

C
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$

E_H DISCHARGED
 $\langle R, h_{\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}} = e$

E_f NEW
 $\langle a, F \rangle_{\mathbb{F}_{2^{128}}} = \mu_f$

OPEN CLAIMS

C
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$

E_f DISCHARGED
 $\langle a, F \rangle_{\mathbb{F}_{2^{128}}} = \mu_f$

LC NEW
 $\sigma = \tilde{a}(r) \tilde{F}(r)$

OPEN CLAIMS

C
 $\rho = \text{Root}(\text{Enc}_C(\text{Pack}_{128}(B_f)))$

LC DISCHARGED
 $\sigma = \tilde{a}(r) \tilde{F}(r)$

E_P NEW
 $\langle B, P \rangle_{\mathbb{F}_{2^{128}}} = \beta$

No extra adapter message, target, or second LinCheck exists. The event rules are now fixed by the PCS profile; **G7** is the remaining implementation and full-fixture closure for this virtual profile.

Step 11. P11/V11 — Open the packed claim

P11. Send the recursive proof and openings. **V11.** Verify them against ρ and exhaust both streams.

For this committed shape, the packed opening target is exactly

$$m_p = t_f + s - 7 = t_f, \quad B : \{0, 1\}^{t_f} \rightarrow \mathbb{F}_{2^{128}}, \quad \text{payload}_{5001} = \text{LE}_{32}(t_f) \parallel \text{Enc}_{\mathbb{F}_{2^{128}}}(\beta). \quad (39)$$

V11 first checks the domain in Equation (39), then absorbs derived tag 0x5001 before the recursive helper. The next tag-0x5002 root must equal ρ . V11 then reuses the PCS ledger without alteration: roots 0x5002/0x5003, sumcheck messages 0x5004, OOD values 0x5005, final vector 0x5006, fold/query nonces 0x5007/0x5008, Merkle Hints 0x5080-0x5082, and squeezes 0x5101-0x5106. Every Hint is authenticated before a later event, and both streams, every nested slice, and the host container must end exactly.

With the still-unresolved partial-core profile named rather than assumed, the composed soundness statement is

$$\varepsilon_{\text{SHA-F2Z}} \leq \frac{d_A + 2d_h}{q} + \varepsilon_{\text{partial}} + \frac{2m_f}{2^{128}} + \varepsilon_{\text{ring}} + \varepsilon_{\text{batch}} + \varepsilon_{\text{open}}. \quad (40)$$

Here $\varepsilon_{\text{partial}}$ includes P6 column compression and P7 GKR; $2m_f/2^{128}$ is the single P9 LinCheck term. Equation (40) is symbolic until **G6** and **G7** publish their proofs and fixtures. P1, P2, P5, and P8 are exact under their stated reject gates.

3 Verifier, status, and gaps

3.1 Verifier algorithm

Algorithm 1 replays P0–P11. The virtual partial core returns Eq. (32); the public adjoint, one LinCheck, and inherited ring switch then follow Eqs. (33), (36) and (38). The direct PCS profile does not enable this virtual path and rejects on Line 4.

Algorithm 1 SHA front-end verifier and virtual-PCS handoff

Require: $\text{pp}_{\text{SHA}}, x_{\text{sha}}, \rho, \pi_{\text{host}}$
Ensure: true or false

- 1: $(p_{\text{core}}, p_{\text{app}}, p_{\text{GKR}}, p_{\text{code}}, g) \leftarrow \text{PROFILEFIELDS}(\text{pp}_{\text{SHA}})$
- 2: **require** $p_{\text{core}} = \text{SHA256-F2Z-VIRTUAL-CORE-V1-00000} \wedge p_{\text{app}} = \text{SHA256-F2Z-MULTICOMP-APP-V1-0000} \wedge \text{ord}(g) = 2^{128} - 1 \wedge \text{PREFLIGHTSHASTATEMENT}(x_{\text{sha}}) = 1$
- 3: $p_{\text{part}} \leftarrow \text{ResolvePartialProfile}(p_{\text{core}}, p_{\text{GKR}}); p_{\text{LC}} \leftarrow \text{ResolveLinCheckProfile}(p_{\text{core}}); p_{\text{ring}} \leftarrow \text{ResolveRingProfile}(p_{\text{core}}); p_{\text{open}} \leftarrow \text{ResolveCodeOpenProfile}(p_{\text{core}}, p_{\text{code}})$
- 4: **require** $p_{\text{part}} \neq \perp \wedge p_{\text{LC}} \neq \perp \wedge p_{\text{ring}} \neq \perp \wedge p_{\text{open}} \neq \perp$
- 5: derive $n_f \leftarrow 6641N + 256, n_h \leftarrow 20005N + 256, m_f \leftarrow \max(22, \lceil \log_2 n_f \rceil), m_h \leftarrow \max(22, \lceil \log_2(n_h + 1) \rceil)$
- 6: $t_f \leftarrow m_f - 7, t_h \leftarrow m_h - 7, n_A \leftarrow 196N + 16, d_A \leftarrow \lceil \log_2 n_A \rceil, d_h \leftarrow m_h$
- 7: **require** $22 \leq m_f \leq m_h \leq 33; b_{\text{host}} \leftarrow \text{PROFILEBOUNDS}(p_{\text{part}}, p_{\text{LC}}, p_{\text{ring}}, p_{\text{open}}, N, m_f, m_h);$ **require** $b_{\text{host}} \neq \perp$
- 8: $(c_N, c_H) \leftarrow \text{DECODEHOSTV1}(\pi_{\text{host}}, b_{\text{host}}, p_{\text{core}})$
- 9: derive canonical $(M_{\text{SHA}}, T, h_0, A_x, \Pi_f); B_x \leftarrow \max_{\beta < 2^{d_A}} \sum_j |A_x[\beta, j]|$
- 10: **require** $\text{SHAANDCENSUS}(M_{\text{SHA}}, T, h_0, A_x, \Pi_f, n_f, n_h, n_A) = 1$
- 11: **require** $2B_x < 2^{39} < q$
- 12: $d_{\text{app}} \leftarrow \text{APPDIGESTV1}(p_{\text{app}}, x_{\text{sha}})$
- 13: $\text{tr} \leftarrow \text{ABSORBPROFILEANDSTATEMENTV1}(p_{\text{core}}, p_{\text{app}}, p_{\text{GKR}}, p_{\text{code}}, d_{\text{app}}, (t_f, 7, 1), g, \rho)$
- 14: $r_{\text{con}} \leftarrow \text{SQUEEZECONSTRAINTPOINT}(\text{tr}, d_A); v \leftarrow \pi_q(A_x)^T \text{eq}(\cdot, r_{\text{con}})$
- 15: $(\text{ok}, \text{tr}, r_h, \tau) \leftarrow \text{VERIFYEVALSHAPESTREAM}(v, \text{tr}, c_N, d_h)$
- 16: **require** $\text{ok} = 1 \wedge |r_h| = d_h; c \leftarrow v(r_h), a_h[z] \leftarrow c \text{eq}(z, r_h)$
- 17: $\mathcal{I}_H \leftarrow (T, h_0, \Pi_f, t_h, 7, r_h, c, a_h, \tau, \rho, m_f, m_h, p_{\text{core}}, p_{\text{GKR}}, g)$
- 18: **require** $\text{VIRTUALPROFILESADMITINSTANCE}(p_{\text{part}}, p_{\text{LC}}, p_{\text{ring}}, \mathcal{I}_H) = 1$
- 19: $\text{tr} \leftarrow \text{ABSORBVIRTUALSEAM}(\text{tr}, 0x1003, t_h, 7, r_h, c, \tau)$
- 20: $(\text{ok}, \text{tr}, R, e) \leftarrow \text{VERIFYVIRTUALPARTIALSTREAM}(p_{\text{part}}, \mathcal{I}_H, \text{tr}, c_N, c_H)$
- 21: **require** $\text{ok} = 1; \mu_f \leftarrow e - \langle R, h_0, \mathbb{F}_{2^{128}} \rangle; \text{Eval}_a(r) \leftarrow \langle R, T_{\mathbb{F}_{2^{128}}}(\Pi_f^{-1} \text{eq}(\cdot, r)) \rangle$
- 22: $(\text{ok}, \text{tr}, B, \beta) \leftarrow \text{VERIFYLINCHECKRINGSTREAM}(p_{\text{LC}}, p_{\text{ring}}, \text{Eval}_a, \mu_f, \rho, \text{tr}, c_N, c_H)$
- 23: **require** $\text{ok} = 1 \wedge \text{dom}(B) = \{0, 1\}^{t_f}$
- 24: $\text{tr} \leftarrow \text{ABSORBOPENINGTARGET}(\text{tr}, 0x5001, \text{LE}_{32}(t_f) \parallel \text{Enc}_{\mathbb{F}_{2^{128}}}(\beta))$
- 25: $\text{ok} \leftarrow \text{VERIFYPACKEDOPENSTREAM}(p_{\text{open}}, \rho, B, \beta, \text{tr}, c_N, c_H)$
- 26: **require** $\text{NOPENDINGHINT} = 1 \wedge \text{EXACTNARGHINTCOUNTSANDBYTES}(c_N, c_H) = 1 \wedge \text{NESTEDCURSORANDHOSTEOF}(\pi_{\text{host}}, c_N, c_H) = 1$
- 27: **return** ok

OPEN CLAIMS

C DISCHARGED
 $\rho = \text{Root}(\text{Enc}_{\mathbb{C}}(\text{Pack}_{128}(B_f)))$

EP DISCHARGED
 $\langle B, P \rangle_{\mathbb{F}_{2^{128}}} = \beta$

3.2 Provenance and current status

Quantity	Provenance and verifier treatment	Status
Public/derived	Validate x_{sha}, ρ , all IDs, g , shape, and d_{app} ; recompute maps and challenges from the bound prefix.	math specified; fixture open
Application stream	NARG round polynomials are checked and absorbed before their dependent challenge.	specified
Core/opening	The partial core is unresolved. The adjoint and one-LinCheck ring contract are specified; their integrated implementation and fixtures are still absent.	open

3.3 Gap register: do not silently resolve in code

	Source omission and this profile's rule	Owner; reviewer	Required closure evidence
G1	FIPS specifies hashing and padding, while this profile proves public chained compressions only. Padding, private blocks, and an empty chain reject.	John Wu; Shreyas Londhe	Public-statement fixtures for one and multiple blocks plus negative scope tests.
G2	The sources do not prove that fixed q is injective on every reachable integer residual. The verifier enforces $q > 2B_x$.	John Wu; Albert	Reviewed symbolic residual bound and executable maximum-row check.
G3	Earlier SHA material omitted feed-forward from its census. This profile includes every feed-forward equation and uses $n_A = 196N + 16$.	John Wu; Albert	FIPS known-answer vectors and an independently recomputed census for several chain lengths.
G4	Prototype paths expose incompatible layouts. This profile has one execution-ordered, LSB-first compact tensor. Nonsemantic filler is committed but unconstrained; honest encoders zero it.	John Wu; Shreyas Londhe	Index-map fixture, honest-zero encoding bytes, and independent root vector.
G5	Freigen proves one compression, but chained row sharing, the padded selector, h -only normalization, and the dual adjoint evaluator remain derived.	John Wu; Albert	Chain equivalence proof and constraint-by-constraint differential tests, including $a(r) = \langle R, T_{\mathbb{F}_{2^{128}}}(\Pi_f^{-1} \text{eq}(\cdot, r)) \rangle$.
G6	P5–P7 have no registered virtual partial-core/GKR ledger or numeric reduction error. The direct PCS profile therefore rejects.	John Wu; Shreyas Londhe	Profile ID, typed reduction proof, byte ledger, numeric error, and positive/negative fixtures.
G7	The PCS now fixes one LinCheck and its reuse of the inherited ring vector, but no integrated implementation or independent full fixture exists. Interoperability is unclaimed.	John Wu; Shreyas Londhe	Two independent verifiers that agree on full proofs and reject round, tag, order, terminal, suffix, root, and opening tampering.